

EEE 443/543 FALL 2025-26 Assignment # 2

Assignment : This assignment is related to curve fitting problem. The following function $h(x)$ is to be approximated by using a multilayer neural network.

$$h(x) = 0.25x^2 \sin(x) \quad -\pi \leq x \leq \pi .$$

1 : Use a 2-layer structure which has two inputs (one representing x , and the other representing the threshold input, i.e. -1), one hidden layer and one output layer. The output layer contains a single neuron with linear activation function, i.e. $f(v) = v$. The number of neurons, n , in the hidden layer is chosen as $n = 10, 15, 20$ and the activation function is chosen as $f(v) = \tanh(v)$.

2 : Choose the training set $\{x(1), x(2), \dots, x(8N + 1)\}$ as

$$x(i) = -\pi + \frac{i-1}{4N}\pi \quad i = 1, 2, \dots, 8N + 1$$

where $N = 20, 30, 40$. Find the desired outputs as

$$d(i) = h(x(i)) \quad i = 1, 2, \dots, 8N + 1.$$

2 : Choose the learning coefficient η as $\eta = 0.1, 0.3, 0.5$. By using Back Propagation (BP) algorithm find the weights so that the average error E_{ave} over the training set satisfies $E_{ave} \leq 10^{-4}$. Record the average error at the end of each epoch and the number of epochs till convergence. Plot E_{ave} vs. epochs. Plot $h(x)$ and $y(x)$ vs. x , where y is the Neural Network output.

Hint : E_{ave} may not decrease monotonically, and the required accuracy may not be achieved. In such a case you may decrease η during BP, e.g. $\eta \leftarrow 0.9\eta$ per a number of epochs, e.g. after each 50 epochs. etc. Also try to minimize the E_{ave} given below. If that does not work as well, you may stop after the minimum error you achieved, or after certain number of epochs, e.g. 1000.

Note that for the training set, E_{ave} is the mean squared error defined as :

$$E_{ave} = \frac{1}{8N+1} \sum_i (d(i) - y(i))^2$$

where $y(i)$ is the output of Neural Network.

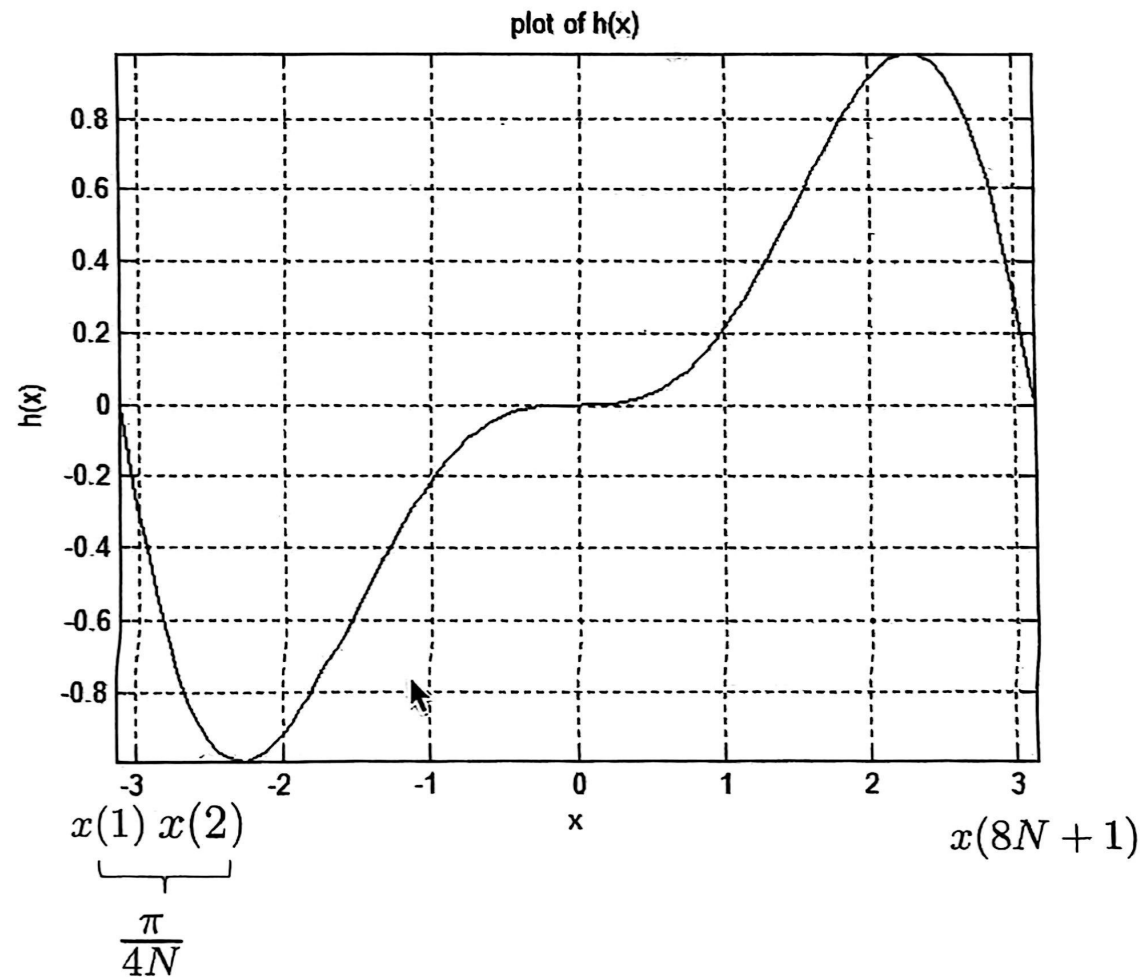
3 : After the training of the neural network, now choose the test set $\{x_{test}(1), x_{test}(2), \dots, x_{test}(8N)\}$ as

$$x_{test}(i) \stackrel{\Delta}{=} \frac{x(i) + x(i+1)}{2} \quad i = 1, 2, \dots, 8N$$

where $x(i)$ is as given in **2**. Find the E_{ave} over the test set.

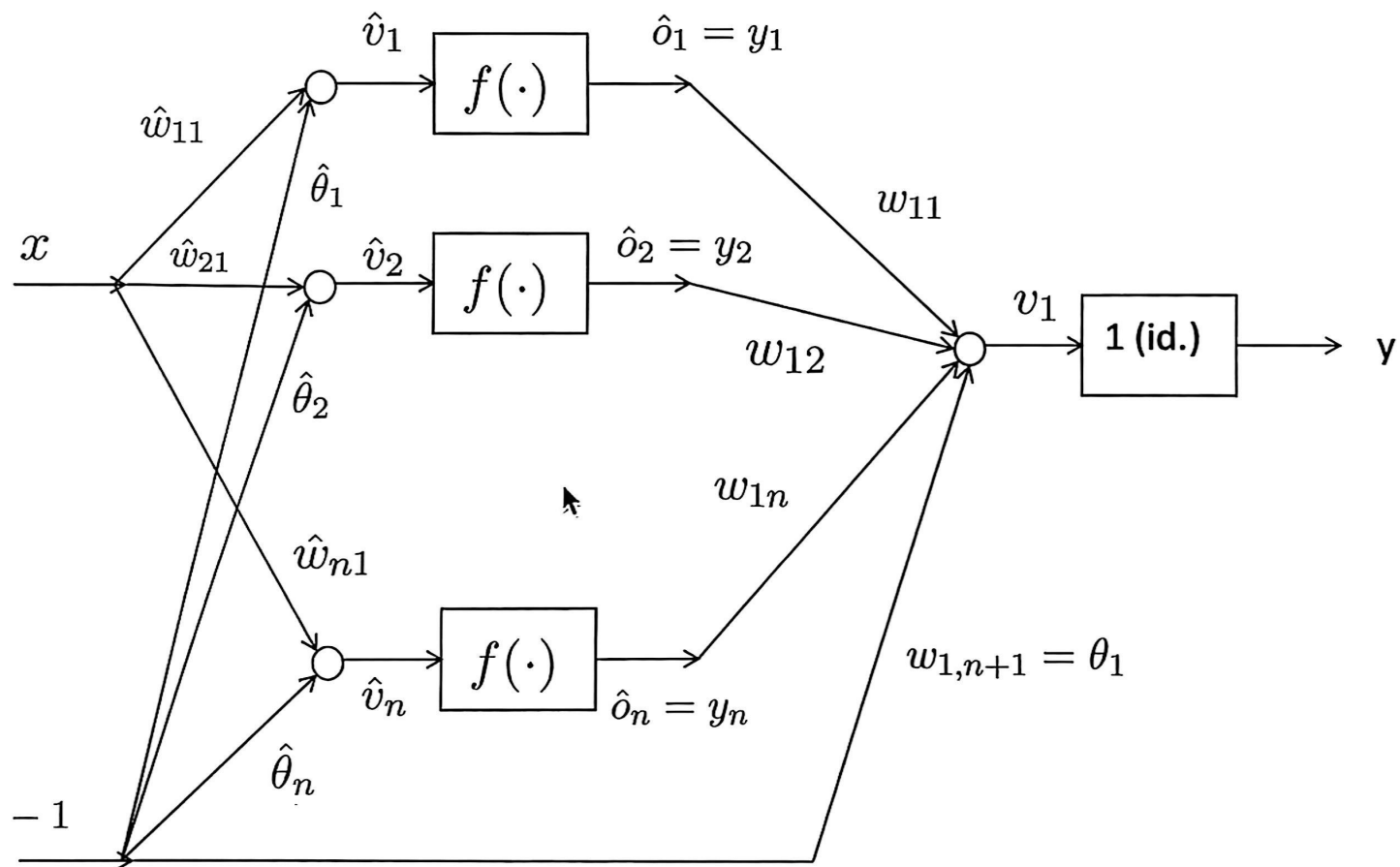
Note : We have three hyper parameters, n , η , N . So you should run the algorithm at least $3 \times 3 \times 3 = 27$ times.

4 : Write your results in a nice report format, preferably in pdf.



$$S_{tr} = \{ \underset{\downarrow}{x(1)} \quad \underset{\downarrow}{x(2)} \quad \dots \quad \underset{\downarrow}{x(8N+1)} \}$$

$$h(x(1)) \quad h(x(2)) \dots h(x(8N+1))$$



$$n = 10, 15, 20$$

Instructions:

1. Prepare a report (including your answers/plots) to be uploaded on Moodle.
 2. The report should be typeset (no handwriting allowed except for lengthy derivations, which may be scanned and embedded into the report).
 3. Show all steps of your work clearly.
 4. Unclear presentation of results will be penalized heavily.
 5. No partial credits for unjustified answers.
 6. **Use of any toolbox or library for neural networks is prohibited.**
 7. Return all ████████ Python code that you wrote in a single ████.py file.
 8. Code should be commented,
 9. The code file should NOT return an error during runtime.
 10. If the code returns an error at any point, the remaining part of your code will not be evaluated (i.e., 0 points).
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